CPEN 416 / EECE 571O

Lecture 04: Projective measurements; introducing multi-qubit systems

Tuesday 16 September 2025

Announcements

$$|0\rangle - \boxed{H} - \boxed{RY(0.2)} - \quad \cdots$$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$RY|0\rangle = \begin{pmatrix} \cos\theta/2 & -\sin\theta/2 \\ \sin\theta/2 & \cos\theta/2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \cos\theta/2\,|0\rangle + \sin\theta/2\,|1\rangle$$

- Quiz 2 today
- Some midterm information to be posted later this week

$$RY(\theta) = \begin{pmatrix} \cos\theta/2 & -\sin\theta/2 \\ \sin\theta/2 & \cos\theta/2 \end{pmatrix}$$

$$RY(\theta)\left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right] = \frac{1}{\sqrt{2}} RY(\theta)|0\rangle + \frac{1}{\sqrt{2}} RY(\theta)|1\rangle$$

$$= \frac{1}{\sqrt{2}}\left[\cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle\right] + \frac{1}{\sqrt{2}} \cdots$$

Last time

We played with *Pauli rotations*.

	Math	Matrix	Code	Special cases (θ)
RZ	$e^{-i\frac{\theta}{2}Z}$	$\begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$	`qml.RZ`	$Z(\pi), S(\pi/2), T(\pi/4)$
RY	$e^{-i\frac{\theta}{2}Y}$	$\begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$	`qml.RY`	$Y(\pi)$
RX	$e^{-i\frac{\theta}{2}X}$	$\begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$	`qml.RX`	$X(\pi), SX(\pi/2)$

Last time

We explored the effect of single-qubit unitary operations on the Bloch sphere.

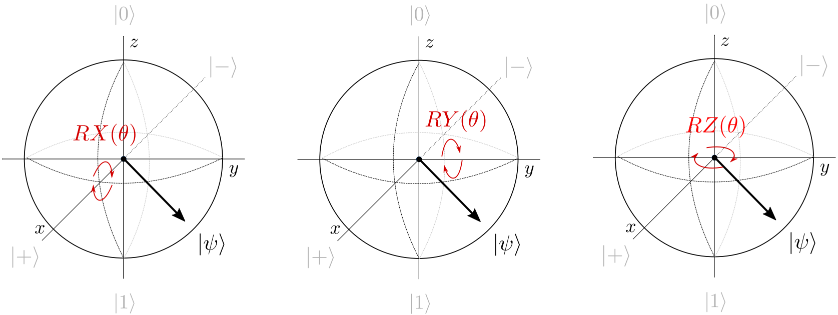


Image credit: Codebook node I.6

Last time

$$(\langle w|v\rangle)^* = \langle v|w\rangle$$

We introduced the “bra” part of braket notation

$$|v\rangle = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \qquad \langle v| = \begin{pmatrix} v_1^* & v_2^* \end{pmatrix}$$

We defined the inner product to quantify *overlap* (similarity) between quantum states.

$$\langle v|w\rangle = v_1^* w_1 + v_2^* w_2$$

toffoli

$$|\psi\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ i/\sqrt{2} \end{pmatrix} \quad |\phi\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} \end{pmatrix}$$

$$\langle\phi|\psi\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} & +i/\sqrt{2} \end{pmatrix}\begin{pmatrix} \frac{1}{\sqrt{2}} \\ i/\sqrt{2} \end{pmatrix} = \frac{1}{2} + \frac{(i)^2}{2} = 0$$

Learning outcomes

1. Perform a projective measurement in the computational basis
2. Apply basis rotations to perform a projective measurement in any basis
3. Mathematically describe a system of multiple qubits
4. Describe the action of common multi-qubit gates

Inner products

Exercise: compute the inner product of the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

with itself.

$$\langle\psi|\psi\rangle = (\alpha^* \ \beta^*)\begin{pmatrix}\alpha \\ \beta\end{pmatrix}$$

$$= \alpha\alpha^* + \beta\beta^*$$

$$= |\alpha|^2 + |\beta|^2$$

$$= 1$$

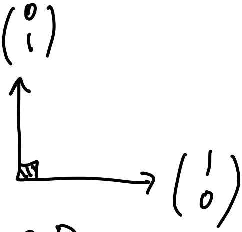
Inner products

Exercise: compute the inner product between all possible combinations of $|0\rangle$ and $|1\rangle$.

$\langle 0\|0\rangle$	1
$\langle 0\|1\rangle$	0
$\langle 1\|0\rangle$	0
$\langle 1\|1\rangle$	1

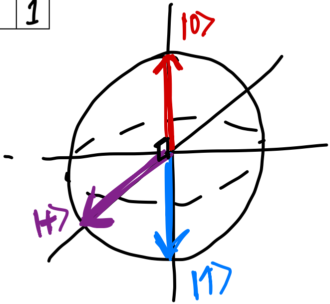
$$\langle +|0\rangle = \frac{1}{\sqrt{2}}$$

$$= \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$



2D

$$\begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0$$



Orthonormal bases

For a single qubit, a pair of states that are **normalized** and **orthogonal** constitute an **orthonormal basis** for the Hilbert space.

Exercise: do the states

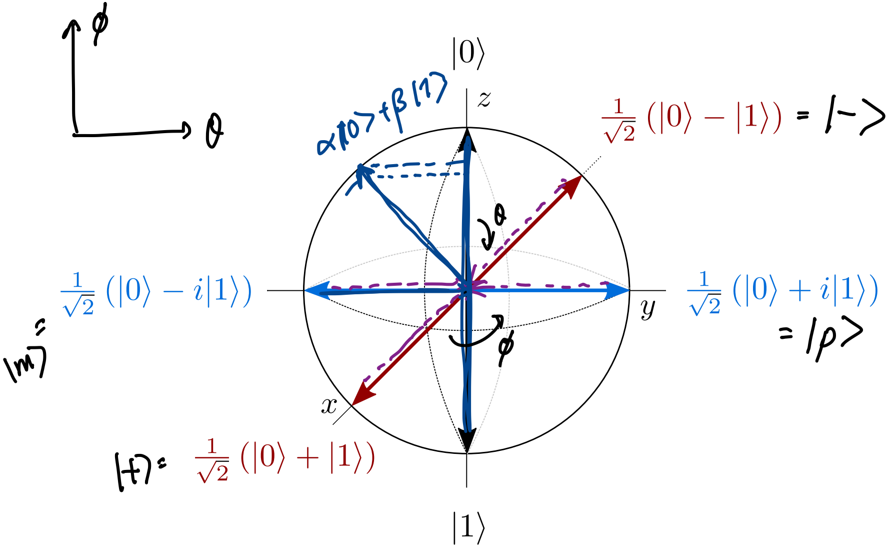
$$|p\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle, \quad |m\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$$

form an orthonormal basis?

$$\langle m | p \rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} & +\frac{i}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix} = \frac{1}{2} + \frac{i^2}{2} = 0$$

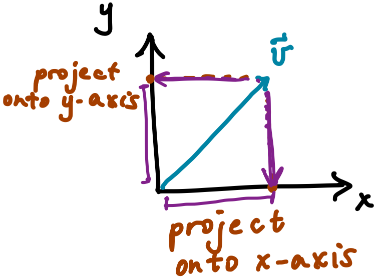
Yes! "Y basis"

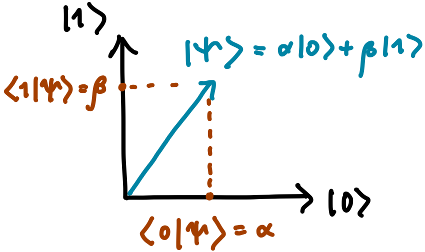
Bases on the Bloch sphere



Projective measurements

Measurement is performed with respect to a basis; we perform **projections** to determine the overlap with a given basis state.





(Image for expository purposes only!)

Image credit: Xanadu Quantum Codebook I.9

Projective measurements

When we measure a qubit in state $|\varphi\rangle$ with respect to basis $\{|\psi_i\rangle\}$, the probability of observing it in state $|\psi_i\rangle$ (outcome i) is

$$Pr(\text{outcome } i) = |\langle\psi_i|\varphi\rangle|^2 \qquad \{|\psi_1\rangle, |\psi_2\rangle\}$$

If we observe outcome i, the qubit *remains in* $|\psi_i\rangle$.

Example: Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. If we measure $|\psi\rangle$ in the computational basis, $\{|0\rangle, |1\rangle\}$

$$Pr(0) = |\langle 0|\psi\rangle|^2 = \left|\begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}\alpha \\ \beta\end{pmatrix}\right|^2 = |\alpha|^2$$

$$Pr(1) = |\langle 1|\psi\rangle|^2 = |\beta|^2$$

Measurement in the computational basis

Exercise: what are the measurement outcome probabilities for

$$|p\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle, \quad |m\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$$

in the computational basis?

$|p\rangle$

$Pr(0) = \frac{1}{2}$

$Pr(1) = \frac{1}{2}$

$|m\rangle$

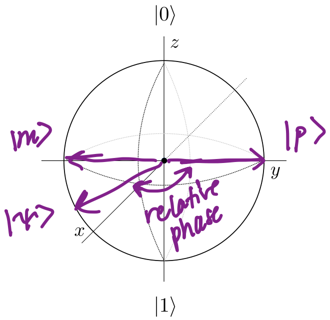
$Pr(0) = \frac{1}{2}$

$Pr(1) = \frac{1}{2}$

Measurement in the computational basis

We know these are *not* the same state: there is a relative phase.

$$|p\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle, \quad |m\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$$



How to tell them apart with a measurement?

Basis rotations

So far we've seen 3 types of measurements in PennyLane:

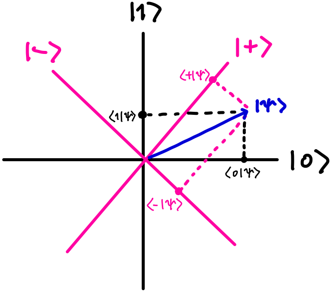
1. `qml.state()`
2. `qml.probs(wires=x)`
3. `qml.sample()`

But these all perform a computational basis measurement. This is also the only measurement allowed on most hardware.

How can we measure with respect to *different bases*?

Basis rotations

Unitary operations preserve length *and* angles between orthonormal quantum states (prove on practice A1!).



For expository purposes only!

Projective measurements can be performed with respect to a different basis by applying a **basis rotation** to remap its states to computational basis states.

Image credit: Codebook node I.9

Basis rotations

Exercise: Suppose we want to measure in the "Y" basis. First, determine a sequence of gates that sends

$$
\begin{aligned}
|0\rangle \quad &\rightarrow \quad |p\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + i|1\rangle\right) \\
|1\rangle \quad &\rightarrow \quad |m\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle - i|1\rangle\right)
\end{aligned}
$$

v1 : –[RX(π/2)]–

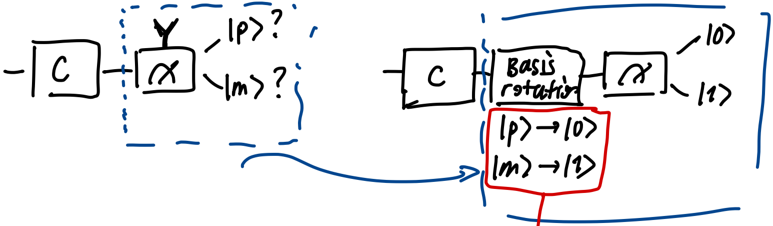
v2 : –[RX(−π/2)]–

v3 : –[H]–[RZ(π/2)]–

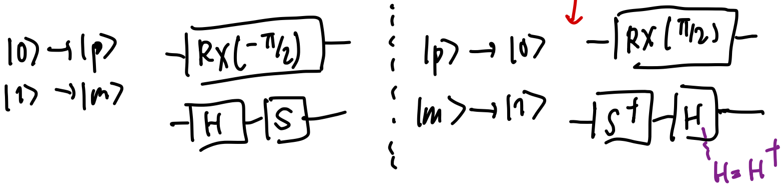
v4 : –[H]–[S]–

Basis rotations

At the end of our circuit, apply the reverse (adjoint) of this transformation to rotate *back* to the computational basis.



If we measure and observe $|0\rangle$, we know the qubit was previously $|p\rangle$ in the Y basis (analogous result for $|m\rangle$).



Adjoints

PennyLane can compute adjoints of operations *and* entire quantum functions using `qml.adjoint`:

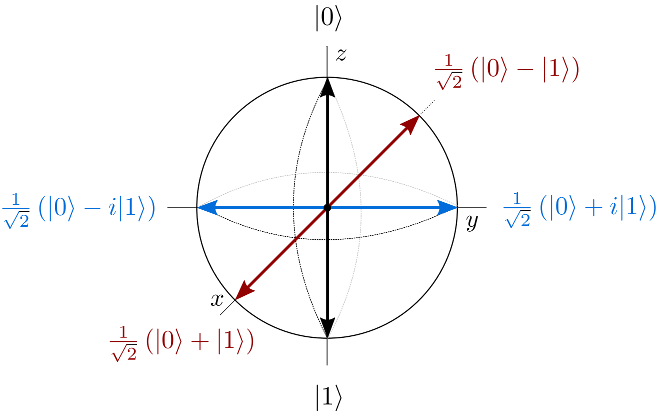
```
def some_function(phi):
    qml.RX(phi, wires=0)

def apply_adjoint(phi):
    qml.adjoint(qml.S)(wires=0)
    qml.adjoint(some_function)(phi)
```

`qml.adjoint` is a special type of function called a **transform**.

Projective measurements and the Bloch sphere

Projective measurement corresponds to a geometric projection onto the axis represented by a basis.



H-RZ exercise, revisited

Exercise: suppose we run the following circuit, but perform the measurement in the Y basis.

1. Using the Bloch sphere as intuition, what do you think the outcome probability of $|p\rangle$ looks like as a function of θ? Discuss with a neighbour.
2. Test your predictions in software.

We'll take up on Thursday.

Next time

Content:

- Entanglement
- Measurement of multi-qubit states
- Bell basis measurements

Action items:

1. Work through last section of yesterday's tutorial (bases)
2. Practice assignment 1 (can do Qs 1, 2, parts of 4)

Recommended reading:

- For today: Codebook IQC, SQ, MQ; N & C 1.2, 1.3.1-1.3.5, 2.2.3-2.2.5, 4.3
- For next time: Codebook SQ MQ (all tied up); N & C 1.3.6, 1.3.7, 2.3